BILL & MELINDA GATES foundation

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INTRODUCTION TO R & PYTHON

DAY 1: 4 May – Introduction to R

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Basic Statistical Analysis

- In this section we will cover the following:
  - Measures of location:
    - Mean & median
  - Measures of spread:
    - Standard deviation & interquartile range
  - Hypothesis testing
  - Correlation & regression analysis

EXAMPLE: South African animals

- Consider again the size and the weight of the birds.
  - For these two variables, calculate the mean, standard deviation, median and interquartile range.
  - Test whether the mean size of the birds is significantly different from 90cm.
  - Test whether the mean weight of the birds is significantly more than 1kg.
  - Test whether the mean log-transformed size of the birds is significantly different from 4.5.

- **Test whether the mean log-transformed weight of the birds is significantly more than 0.**

- **Calculate the correlation coefficient between the weight and the size of the birds.**

- **Calculate the correlation coefficient between the log-transformed weight and the log-transformed size of the birds.**

- **Fit a simple linear regression model in which the log-transformed weight of the birds is explained by the log-transformed size of the birds.**

- **Use the fitted linear regression model to predict the weight of the arctic tern.**

EXERCISE: Arachnophobia

- Consider again the study on arachnophobia.

- Variables:

  y_i: The GSR measurement for the level of anxiety of the i^{th} arachnophobe.

  x_i: The size of the spider in centimeters (cm) for the i^{th} arachnophobe.

  - For these two variables, calculate the mean, standard deviation, median and interquartile range.

- **Test whether the mean size of the spiders is significantly different from 10cm.**

- **Test whether the mean GSR measurement is significantly more than 25.**

- **Calculate the correlation coefficient between the GSR measurements and the sizes of the spiders.**

- **Fit a simple linear regression model in which the GSR measurements are explained by the sizes of the spiders.**

- **Use the fitted linear regression model to predict the GSR measurement for Nosnow Cannotski who had to interact with a spider of 13cm.**